

Unit 2: Models of E-Governance (12 Hrs.)

Models of Digital Governance

It is clear that e-governance is defined fundamentally as applications of ICT to governance activity. However, this can be manifested in multifarious ways and models. Models for e-governance, especially in the developing countries, are essential for a right perspective on e-governance implementation.

Digital governance in developing countries may seem to be a distant thought for many but it is certainly making its presence felt, and in a form which is different from that evident in the developed countries. The growing use of information and communication technology (ICT) is catalyzing the formation of knowledge societies, and thus providing greater avenues to people to participate in their own development process.

The inhabitants of knowledge societies will be more informed about the socio-economic and political processes which affect them and will be empowered to voice their concerns, and make informed decision on how they would like to be governed and by whom.

We are entering into a brave new world where it will not be the leaders who govern people but it is the people who will let the leaders govern them.

Models of digital governance are still evolving in developing countries. A few generic models have been shaped up, which are finding greater recognition and are being replicated. These models are based on the inherent characteristics of ICT such as enabling equal access to information to anyone who is a part of the digital network and de-concentration of information across the entire digital network, connecting all sources of information. In simpler terms, information does not reside at any one particular node in the Digital Governance Models but flows equally across all the nodes, a fundamental change from the more common hierarchical information flow model that leads to unequal distribution of information and hence skewed power relations.

Hierarchy is inherent in the government departments. Equity based information flow may not be always compatible with government functioning. Therefore, appropriate administrative reforms and some

reengineering may be required before e-governance may be really implemented.

It needs to be noted here that these models of governance are fundamentally different from those which are popular in developed countries due to differences in the basic conditions, and perspectives and expectations from good governance. The six generic models of digital governance in developing countries are:

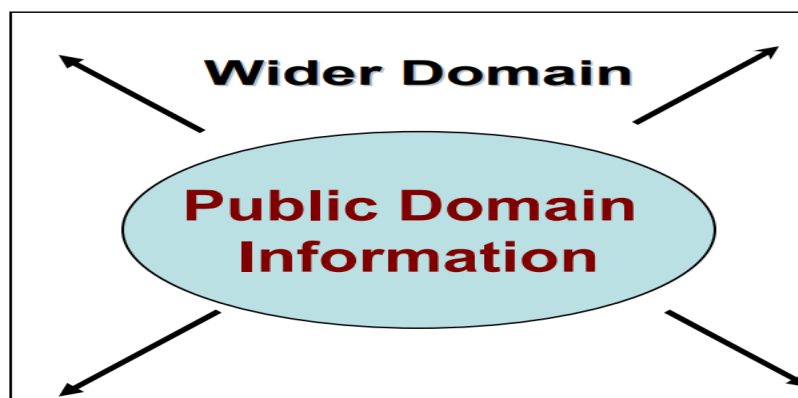
1. Broadcasting/Wider Dissemination Model
2. Critical Flow Model
3. Comparative Analysis Model
4. Mobilization and Lobbying Model
5. Interactive Service Model

These models exhibit several variations dependent on the local situation and the governance functions carried out through these models

1. Broadcasting/Wider Dissemination Model

Principle

The model is based on dissemination of information relevant to better governance that is already in the public domain into wider public domain through the use of ICT and convergent media.



The rationale behind the model is that a more informed citizenry is able to understand better the governance mechanisms and is more empowered to make informed choices and exercise its rights and responsibilities. Further, there is a greater likelihood that the society in which the individuals are equally informed will ensure that the agenda and forms of governance are not biased to favor few.

The wider-dissemination model opens up an alternate channel for people to access information as well as validate information available in the local domain from external sources. The widespread application of this model gradually corrects the situation of information failure and provides people with the basic government-related information to come to a common understanding and decide upon the future course of action.

Applications

1. Putting government laws and legislation online.
2. Making available the names, contact addresses, e-mails, and fax numbers of local governmental officials online.
3. Making available key information pertaining to governmental plans, budgets, expenditures, and performances online
4. Putting key court judgment/judicial statements that are of value to common citizens and creating precedence for future actions online, viz. key environment related judgments, State Vs Citizen court rulings, etc.

Project on GISTNIC (General Information Services Terminal of National Informatics Center) is an example of this model. In this project, the government agency (NIC) disseminates general information of about 25 subjects such as Economy, Education, Census, Tourism, etc. to general public. Government Orders (GOs) also are being publicized. However, after the advent and popularity of Internet, almost all government departments have been setting up or maintaining websites providing information about themselves to the public general.

Evaluation

This model is the first step to more evolved forms of digital governance models. It is also the most crucial one as it catalyzes free access and flow of information to all segments of society and serves as the building block to better governance.

The model, however, loses its effectiveness where free flow of information is not encouraged or it is not an objective. Tight governmental controls and bids to censor the content being transmitted through this model would be the bane of the model. The responsibility is therefore both on governmental

organizations as well as civil society organizations to ensure that such models continue to proliferate.

Organisations / Projects based on the model

India

Directory of official websites of Government of India: <http://goidirectory.nic.in/ministry.htm> National Informatics Centre (India) is the official website of the Government of India. It makes available information on government ministries- its projects and schemes, Indian laws and legislation, contact details of local government offices and key position holders online for public access.

Brazil

Brazil's official national E-Government website: <http://www.gov.br> The website provides comprehensive information on Brazilian government as well as links to integrated citizen services.

South Africa

Chapter 2 Network: <http://www.advocacy.org.za> The Chapter 2 Network is a clearinghouse of information and communication for social justice issues in South Africa. It provides information about advocacy campaigns, research on political intelligence, policy analysis and legislation monitoring to civil society organizations engaged in social justice advocacy.

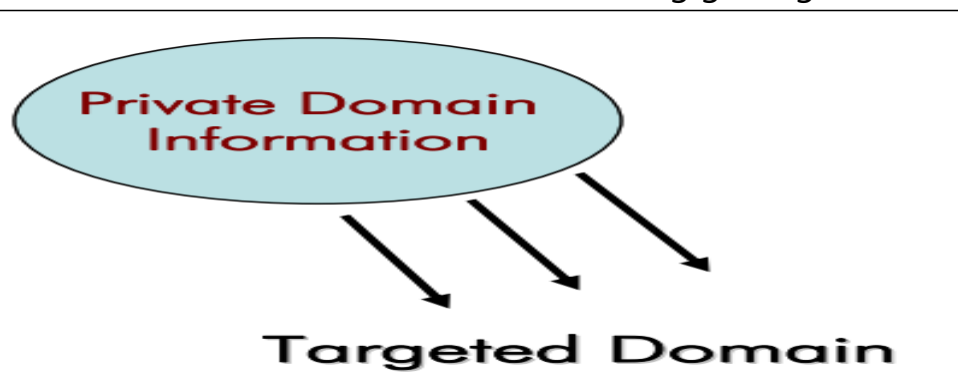
Global

Earth Negotiations Bulletin: <http://www.iisd.ca/voltoc.html> A reporting service that keeps citizens around the world informed about global environmental negotiations, processes and decisions. It has immense value for people and government officials alike in developing countries to keep track of global negotiations taking place in the West and be more informed about them.

2. Critical Flow Model

The model is based on channeling information of critical value to a targeted audience or spreading it in the wider public domain through the use of ICT and convergent media. The model requires foresight to understand the significance of a particular information set and use it strategically. It may also

involve locating users to whom the availability of a particular information set would make a critical difference in initiating good governance.



The strength of Critical Flow model is the inherent characteristics of ICT that makes the concept of distance and time redundant. This reduces the cases of exploitative governance possible earlier due to time lag between availability of information to different users.

Applications

The applications involve making available

1. Information on corruption (by an appropriate legal authority) of a particular government ministry or government officials, to its electoral constituency or to the concerned governing body (e.g. the web sites of Central Vigilance Commission);
2. research studies, enquiry reports and appraisals commissioned by the government to the affected parties;
3. human rights violation and criminal impeachment records against government officials to NGOs and concerned citizens; and
4. environment related information to local communities, for example, information on radioactivity spills, overflow discharge in rivers, green ratings of a company, etc.

Evaluation

Critical Flow Model is more focused in terms of its information content and its intended users. Due to critical aspect of information, the model exposes the weakest aspects of governance and decision-making mechanisms and informs people about specific cases of state failure and bad governance to build up a case for concerted action. At the same time, by fuelling public unrest, the model exerts pressure on the concerned government institutions

and individuals to take into awareness the interest and opinion of the masses in decision making processes. The responsibility of creating such models may lie more with the civil society organizations to emerge as an effective watch to guard to government policies and actions. The model will not work in cases where government mechanisms do not foster public debates and censure all information of critical nature. It will also fail where the government maintains a tight control over all information. There is remains restricted to top few levels of the government. Inherently the Internet is an open medium. Thus restricted dissemination is only typical—only those interested may use the critical and subject based information lodged on Internet web sites for public access as applications.

Organizations / Projects based on the model

India

Central Vigilance Committee (India): <http://cvc.nic.in> An initiative on e-vigilance. The website provides free-access information to citizens about government officials who have been indicted on judicial charges relating to corruption and have been advised penalty. People can also file complaints against any public servant who fall within the jurisdiction of the Commission.

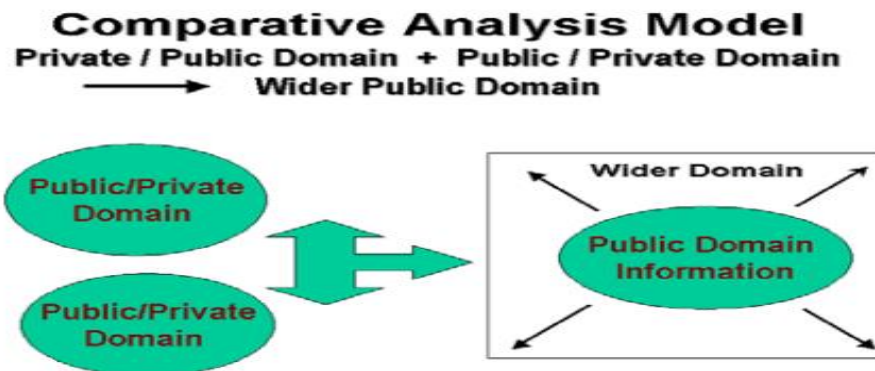
Bangladesh

Human Rights Portal: <http://www.banglarights.net> The Bangladesh Human Rights Network actively promotes human rights reforms both within Bangladesh and across geographical and political boundaries, and supports women, children, and marginalized communities in resisting social oppression.

Global

Transparency International's Corruption News:
http://www.transparency.org/press_moni.html. A trial service run by Transparency International called the "The Daily Corruption News" which reports on corruption from around the world.

3. Comparative Analysis Model



This is one of the least-used but a highly effective model that is gradually gaining popularity. Comparative Analysis model is based on exploring information available in the public or private domain and comparing it with the known information sets to derive strategic learnings and arguments. The model continuously integrates new knowledge products and uses them as a benchmark to evaluate, influence or advocate changes in current governance policies and actions. The comparison could be made over a time scale to get a snapshot of the past and present situation (before-after analysis) or between two different situations to understand the effectiveness of an intervention. (with-without analysis). The strength of this model lies in the boundless capacity of ICT to store information in a retrievable manner and transmit it almost instantaneously across all geographical and hierarchical barriers.

Applications

Some of the situations in which the model could be used are:

- For evaluating the effectiveness of current policies by collecting learnings from government policies and actions in the past.
- Establishing conditions of prior-precedence, especially in the case of judicial or legal decision-making and using it to influence future decision-making. This could be useful in resolving patent-related disputes, public goods ownership rights etc.
- Enabling informed decision-making at all levels by enhancing the background knowledge and providing a rationale for future course of action.

- Evaluating the performance record of a particular government official or ministry.

Developing countries can effectively use this model to their advantage as ICT opens access to global and local knowledge products at a relatively low cost. Watchguard organizations and monitor-groups could use the model to track the performance records of electoral candidates and share them in their constituency. The model is however needful on the availability of comparative information sets and the ability of users to analyze and bring out strong arguments or self-explanatory graphics from the analysis. The model becomes ineffective in absence of a strong civil society interest and short public memory.

Organisations / Projects based on the model

India

Comparative Learning from Disasters

After the devastating earthquake in Kutch in 2001, learnings on disaster management was drawn by comparing it with the high intensity earthquake that shook Latur in India in 1993. The extensive information available on Internet on both these earthquakes open up vast scope of comparison by all segments of the society. An example of disaster comparison made by an Officer of the Indian Administrative Service (IAS) is available at <http://www.cddc.vt.edu/digitalgov/Latur-Gujarat.htm>.

Comparing disaster information can empower people by making them realize that damages to life and property incurred are not just a factor of magnitude of the disaster but are also dependent on the preparedness of the government machinery and effective Government policies to handle the disaster.

India

Green Ratings Project:

<http://www.oneworld.org/cse/html/eyou/eyou31.htm> The Centre for Science and Environment in India conducts a survey of how Green the Indian industries are. The Green Rating Project is an attempt to provide an independent and fair evaluation of the comparative environmental performance of companies, from a perspective which supports

responsible industry and encourages poor performers to improve. Performance reports of companies are shared over Internet and other media and people's opinion is solicited in deciding the greenest environmental managers, Chief Minister etc.

Philippines

Vote.ph: <http://www.vote.ph>

Vote.ph is the non-partisan, non-sectoral online directory and quick reference center on Philippine elected government officials and election candidates for both the national and the local levels. It serves the purpose of informing voters know who are the electoral candidates running in their respective area and who their incumbent elected officials are.

Global

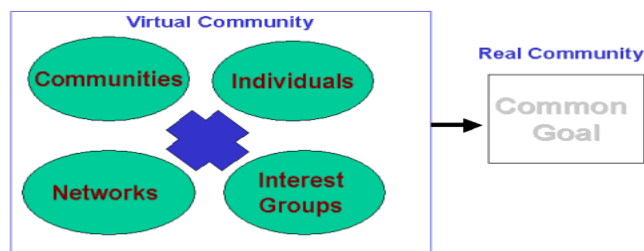
Human Development Indicators

<http://www.undp.org/hdro/indicators.html>

The Human Development Report of UNDP makes use of archived statistical information pertaining to literacy, health, national income etc. as a benchmark to assess the progress made by different countries with regards to their Human Development Index and suggests policy recommendations based on that.

4. Mobilization and Lobbying Model

Mobilization and Lobbying Model is one of the most frequently used digital governance model and has often come to the aid of the civil society organizations in developing countries to impact international decision making processes. The model is based on planned, directed, strategic flow of information to build strong virtual allies to strengthen action in the real world. It takes up the pro-active approach of forming virtual communities which share similar values and concerns, promoting active sharing of information between these communities, and linking them with real world activities.



The strength of this model is in the diversity of its virtual community, and the ideas, expertise and resources accumulated through virtual forms of networking. The model is able to effectively overcome geographical, institutional and bureaucratic barriers to shape concerted action. The model also provides a strong virtual arm to several activities such as directing campaigns against a particular individual or decision-making body.

Applications

Some of the examples in which the model could be used are:

- Fostering public debates on global issues, themes of upcoming conferences, treaties etc.
- Formation of pressure groups to pressurize decision-makers to take their common concerns into cognizance.
- Amplifying the voices of marginalized groups who are traditionally marginalized from decision-making process.
- Building up wider participation in decision-making processes.
- Building up global expertise on a particular theme in absence of localized information to aid decision-making.

The model has grown tremendously since the onset of debates for the Seattle round of World Trade Organization (WTO) in 1999 when it saw the formation of several virtual communities to advocate the concerns of developing countries in the WTO agreement. The display of a unified civil society force at Seattle was in many ways a result of intensive discussions that took place over virtual networks months prior to the summit. The discussions taking place over the virtual network fed into regional level action plans that built into the global movement.

The mobilization and lobbying model enhances the scope of participation of individuals and communities in policy issues and debates. The model also creates an effective warning for government bodies and individuals to be watchful in their actions in case they turn the opinion of local and global community against them. This model could be effectively used by the Government to encourage public debates and to gauge public opinion on a particular issue as a part of good governance strategies.

Organisations / Projects based on such models

Global

Independent Media Centre: <http://www.indymedia.org/> The Center was established by various independent and alternative media organizations to provide grassroots coverage of WTO-Seattle in 1999. The Center acted as a clearinghouse of information for journalists, and provided up-to-the-minute reports, photos, audio and video footage through its website.

Global

GreenpeaceCyber-activistCommunity : The Greenpeace Cyber activist Community has members from more than 170 countries and territories. and is growing quickly. It is an effort towards creation of virtual communities to mobilize global support against some of the disputable environmental policies/ actions of the Government. Members receive email campaign updates, and can send letters to key corporate and political decision makers, download action kits, send e-cards to friends and colleagues, participate in on-line discussions, and play a significant role in helping to win environmental campaigns.

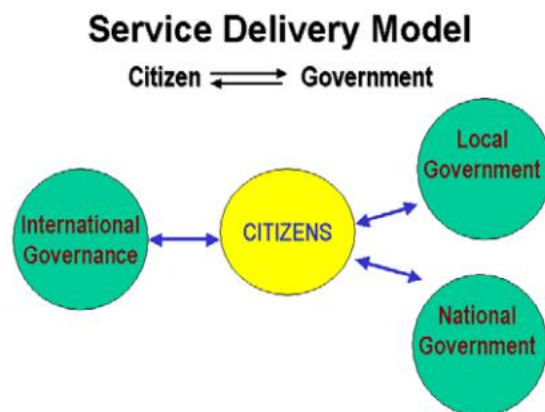
India

Panchayats:

A discussion list run by Indian professionals to discuss policy issues and case-studies relating to local livelihood strategies and decentralized natural resources management. The participation is open and the list generates useful discussion between practitioners, activists and policy makers.

5. Interactive-Service Model / Government to Citizen to Government Model (G2C2G)

Interactive-Service model in many ways is a consolidation of the earlier digital governance models and opens up avenues for direct participation of individuals in the governance processes. ICT as mentioned earlier have the potential to include every individual within a knowledge network and enable interactive communication channels among them. This model fully captures the potential of ICT and leveraged it for greater participation, efficiency and transparency in functioning of the government as well as savings in time and costs relating to decision-making.



The Interactive-Service Model makes possible the various services offered by the Government to be directly accessible to the citizens. It creates an interactive Government to Consumer to Government (G2C2G) channel in various functions such as election of government officials (e-ballots); filing of tax returns, procurement of government services, sharing of concerns and providing expertise; conducting opinion polls on public issues, and grievance redressal.

This model is more embedded in developed countries and has often been proposed for replication in developing countries. Such forms of solution-transfers may not be very effective. The model is on the higher end of technology-reliance as compared to the other models. This

makes it difficult to replicate in developing countries in absence of individual and secure ICT access.

Applications

Some of the situations in which the model could be used are:

- Establishing an interactive communication channel with policy-makers such as video conferencing and online dialoguing.
- Conducting electronic ballots for the election of government officials and other office bearers.
- Conducting public debates / opinion polls on issues of wider concern before formulation of policies and legislative frameworks.
- Filing of grievances, feedback and reports by citizens with the concerned governmental body.
- Performing governance functions online such as revenue collection, filing of taxes, governmental procurement, payment transfers etc.
- Carrying out video-conferencing, on-line discussion with policy makers.

Organisations / Projects based on the model

Philippine

Philippine Custom Reform:

<http://www1.worldbank.org/publicsector/egov/philippinecustomscs.htm> The Philippines Customs Bureau has developed an on-line system to process clearance of imports, payment of duty, and delivery of release orders for shipments to leave the docks. The new on-line system has reduced the cost of trade for businesses, reduced opportunities for fraud, and helped the Bureau to maximize revenue collection.

India

Gyandoot: <http://www.gyandoot.net/gyandoot/intranet.html>

Gyandoot is an intranet in Dhar district of Madhya Pradesh connecting rural cyber cafes catering to everyday needs of the masses. The site offers several interactive governance related services to the local people such as providing copies of land-maps, online registration of applications, and public grievance redressal. It is a step towards tele-centre based e-governance models.

Evolution in E-governance and Maturity Models

The E-governance Maturity Model (EMM—version 1.0) based on the conventional software maturity models, proposes some levels of maturity, depending on the effectiveness with which e-governance efforts have been initiated, implemented or successfully completed. Traditionally, e-governance has been defined as an ICT enabled route to good governance with a view to enhancing transparency in the system and provide prompt and quality services to the citizen. E-governance is an evolutionary path, whose effective implementation requires a complete understanding of its various elements and at the same time taking a holistic view to say focused on its overall objectives.

E-governance journey encounters several milestones that need to be identified and modeled so that efforts invested can be assessed and an appropriate course of action be taken by the organization to further its way on the path of e-governance. The E-governance Maturity Model (EMM—version 1.0) proposes five levels of maturity, depending on the effectiveness with which the e-governance efforts have been initiated, pursued , utilized and institutionalized. EMM 1.0 will facilitate government organizations to assess the current level of e-governance initiatives and accordingly make efforts for the future. The model further identifies the characteristics exhibited by organizations at various levels of maturity that will facilitate correct assessment of the current status. The model also provides Key Focus Areas (KFAs) that need to be focused on to attain a particular maturity level.

Five Maturity Levels

The E-governance Maturity Model (EMM—version 1.0) is based on the fact that speed, openness and ubiquity are some of the major capabilities of ICTs, which can be leveraged for generating transparency, responsiveness and accountability. In the system, on the one hand and empowering the common man by providing faster access to right information at the right time, on the other. It is also based on a service-oriented approach, where public administration is seen as a professional activity and efficient delivery of services to the

internal and external users (customers) is emphasized as a key performance indicator of the government department. The internal customers/users of an organization are its employees and the external users are the citizens, businesses, other government and non-governmental bodies that the organization needs to satisfy in its e-governance endeavor. However, there may exist organizations that are performing well even without ICT application to its functions. Even such organizations may gradually take up to ICT for the sake of other reasons and other benefits. The maturity levels, described below, provide a necessary mechanism to benchmark the efforts invested by an organization of its customers/users.

Level 1: Closed

This is the stage when an organizations does not use ICT as a facilitator for good governance and has no plans to do so in the near future. This situation may arise due to lack of exposure to ICTs and associated benefits that again may depend upon a number of reasons; remoteness from the mainstream in terms of location is primary, and lack of resources and strategic thinking could be some of the other issues. As a result, the organization is 'closed' in terms of being connected and sharing of information in the context of 'e'-governance. However, even in this condition the organization may be efficiently functioning. Given the trend today, all organizations may take up ICT implementations sooner or later.

Level 2: Initial

This level corresponds to the stage when an organization has initiated the automation of its processes but on an ad-hoc basis. No organized efforts are made to undertake the e-governance initiatives. Also, due to lack of direction many such e-governance efforts are abandoned at a subsequent date.

Level 3: Planned

The e-governance initiative, at this level, is undertaken with a systematic approach. The organization has a clearly defined vision, objectives and goals for e-governance. A need assessment study is conducted to prioritize areas of implementation and gauge the extent of e-readiness. Taking input from the need assessment study, extensive planning has been carried out indicating policies, strategies, various

activities, stakeholders, roles and responsibilities and resources required in terms of time, money and manpower to undertake the e-governance exercise. However, the organization is yet to enter into the planned implementation of the e-governance exercise, even though all the requisite planning is completed.

Level 4: Realized

This level corresponds to the stage when the organization actually realizes the complete e-governance plan. Consequently, an integrated system is established where all the internal processes of the organization are computerized and there is seamless information exchange among all concerned entities. The organization starts delivering the services to its external as well as internal customers/users in an effective manner. Complete realization of the plan, in a single instance, would require enormous amount of resources in terms of time, money and manpower, which may necessitate adopting a phases approach for operationalizing the e-governance services. Accordingly, a further classification within this level has been proposed that measures the extent of realization of the plan over a period of time. These sub-levels are also indicative of the openness and effectiveness with which the information is exchanged among the various entities of the organization (external and internal).

- (a) Retrospected. At this level, the organization has retrospectively studied its business processes in view of its vision, overall e-governance objectives, the series-oriented approach (wherein the government is expected to effectively deliver the services to its customers/users), and changes, if required, in the processes are initiated as a constant evolutionary process.
- (b) E-ready. In this stage, e-readiness essentials, which are also the building blocks for e-governance, are ensured by the organization.
- (c) Partially open. At this stage some of the e-governance services are operationalized resulting in a partial information exchange among the entities, both within and outside the organization.
- (d) Open. This sub-level of realized state implies complete deployment of e-governance services that ensure an integrated system that is open to information exchange. The focus here shifts from acquiring and implementing 'e' enabling factors to effectiveness with which the

services are delivered. The system gains responsiveness to deal with the customer needs and is accountable for its services.

Level 5: Institutionalized

At this level, the organizations sustain the realized state over a period of time so that e-governance becomes part of its work culture. The e-governance services are effectively utilized and accepted by the users. Several iterations between planned and realized state lead to institutionalization, when e-governance becomes a way of life.

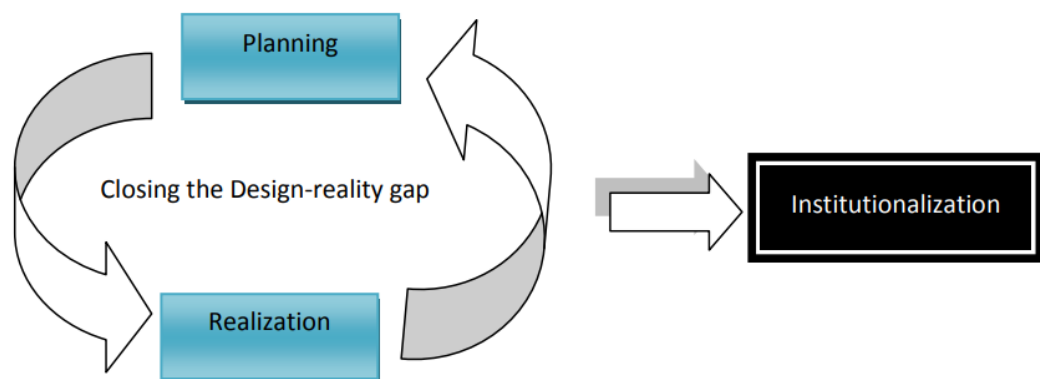


Figure: Reaching the Institutionalized Stage

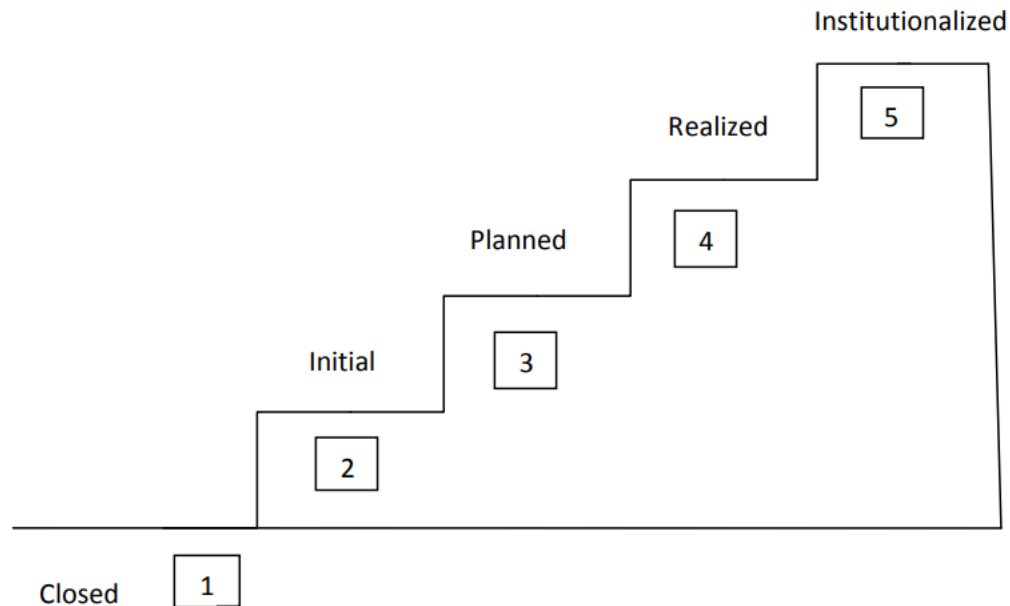
Following define the e-readiness of an organization:

- (a) Presence of strategic thinking, leadership and commitment among top-level decision makers. It is necessary to have visionaries at the top level who can put e-governance on the agenda of the organization and are able to carry it through to implementation and effective utilization.
- (b) Institutional infrastructure. There should also be institutions responsible for creating e-governance awareness among stakeholders and users, and coordinate the e-governance exercise.
- (c) ICT infrastructure. A sound computing and communication infrastructure is an essential requirement for effective operationalization of e-governance services.
- (d) Human capacities. It is important to build human capacities in terms of necessary knowledge and skills to initiate, implement and sustain e-governance initiatives. It is equally important to generate right attitude that is receptive to ICT based administration and ICT based delivery of services. All require extensive and intensive training and ICT orientation

programs at various levels of the staff and officers in the concerned government organization.

(e) Legal Infrastructure. The necessary laws and regulations should also be in place to support ICT as a tool for good governance.

E-governance maturity levels can be depicted as in the figure below:



Characteristics of Maturity Levels

The characteristics exhibited by the organizations at various levels of maturity facilitate assessing the current level of e-governance attained by the organization. The following points describe the characteristics of the organizations at various levels

Level 1: Closed

- Organizations are closed to e-governance. No plans or vision is available. They continue with fully manual and conventional operations.

Level 2: Initial

- Organization lacks strategic thinking and direction for e-governance at top level.
- There are unorganized and isolated efforts of automation in some areas.
- Automation efforts are a result of individual's initiatives.

- The effort sustains as long as the enthusiasm of the individual, who initiated it, remains and is often abandoned due to lack of direction.
- Such organizations generally accumulate hardware without any planning and much of it goes unutilized or underutilized.

Level 3: Planned

- E-governance is a part of organization's agenda.
- Strategic thinking and leadership guide the e-governance initiatives.
- Clear understanding of e-governance needs as projected by the external and internal customers/users.
- Extensive plan is prepared for implementing e-governance, addressing all Key Focus Areas (KFAs) and other related issues.
- All the necessary documents for e-governance planning are in place. These documents include Vision and Scope document for e-governance, Need Assessment Survey document, Policy guidelines, and Action Plan and Outsourcing guidelines.

Level 4: Realized

Retrospected.

- Business processes are attuned with the vision and overall e-governance objectives.
- There is awareness about e-governance among all concerned—the stakeholders as well as the customers/users.

E-ready.

- The organization has a sound infrastructure (technological, institutional, legal, and human) in place, for implementing e-governance.
- The customers/users are oriented and motivated to use e-governance services.

Partially open.

- Some of the e-governance services are deployed, leading to partial information exchange among the entities.
- Partially open organizations sometimes focus only on their internal or backend processes, allowing an information exchange within the limits of the organization thus remaining insulated from its external entities. In such cases, Government-to-

Employee (G2E) interface is visible, whereas Government-to-Citizen (G2C), Government-to-Government (G2G) and Government-to-Business (G2B) interfaces are not yet established.

- Another case of partial deployment of e-governance services result in a visible G2C interface with no emphasis on building G2E, G2B, or G2E interfaces. In an enthusiasm to quickly open up its front-end, the organization negligibly focuses on computerization of the supporting backend processes, thus creating hollowness behind the face of static web sites.
- A hybrid of the above two cases is characterized by building interfaces with external entities and simultaneously focusing on computerizing the corresponding backend processes, thus opening a balanced information exchange among the internal as well as the external customers of the organization.

Open.

- The organization has an integrated system, reflective of smooth information exchange within and outside the organization, i.e. Government-to-Employee (G2E), Government-to-Citizen (G2C), Government-to-Government (G2G) and Government-to-Business (G2B) interfaces are well established over a sound foundation of e-governance building blocks (the e-readiness essentials).
- Organization focuses on satisfying the users of e-governance services.
- The internal and external customers/users of the organization start utilizing the e-governance services and become dependent on them.

Level 5: Institutionalized

- The e-governance system of the organization is driven by a well established Knowledge Management System that generates an ability in the organization to evolve with time in view of new requirements.
- E-governance becomes an effortless exercise for the organization and it becomes a way of life for the stakeholders and customers/users.

- The organization at this level is completely paperless.

Towards Good Governance through E-Governance

The digital governance models bring about a transformation in the existing forms of governance as they change the nature of citizen-governance relationship and bring in new agents and mechanisms to influence the governance processes. They foster democratic control over the governments' economic, social and welfare policies by citizens and civil society organizations—a key process requirement for good and responsive governance.

The changes brought about in the citizen-governance relationship through digital governance are fourfold:

1. They open up avenues for flow of information both vertically and laterally to involve a wider foundation of the civil society. A greater density of information flow is achieved between government and civil society, or within civil society itself. The right to voice and expression gets gradually embedded among citizens through digital means.
2. Information becomes difficult to be capitalized by a few for political gains at the expense of ignorance of citizens. The power equations shift from being concentrated at select nodes to its even distributions among citizens, opposition parties and watchdog groups.
3. There is a greater scope influence policy-makers and members of civil society through collective opinion, direct participation, participation in public debates and use of advocacy tools.
4. Policy-makers become more aware of the voices of people and can effectively involve them in policy-making mechanisms. They realize that their actions are under the scrutiny of many more watchdog organizations and there are greater avenues available with people to obtain any information. Information also becomes difficult to demolish and is forever archived to increase the institutional memory of society.

It may, however, be noted that all these features described above are applicable only to well established democracies with sound systems in place.

This criterion may not be applicable to many developing countries and even some developed countries.

The widespread application of digital governance models synergizes representative forms of democracy with direct participatory forms. This becomes possible because information earlier residing with citizen's representatives in the governance domain is now available with the citizens themselves. People are therefore more aware of the political issues which interests them, and also about the implication of the actions made by their representatives.

This form of informational equality creates an effective watchdog system where people watch those who are supposed to guard their interests in the governance mechanisms. This can potentially catalyze popular democracy at local, regional, and national levels. The following table summarizes the nature of people's participation in e-governance.

Table: People's Participation in ICT-Enabled Governance

	<i>Conventional Media</i>	<i>ICT and Convergence Media</i>
Mode of participation	Representative of ex-situ	Individual/Collective in-situ
Form of participation	Passive/Reactive	Pro-Active/Interactive
Impact of Participation	Indirect	Direct/Immediate

As seen from the comparison chart, the involvement of people in ICT-enabled governance mechanisms is significantly different in comparison to traditional forms of governance. The mode of governance transforms from "representative" to "individual based" and from "passive" to "proactive". People are no longer totally dependent on information provided to them by their leaders or policy makers through conventional media to form an opinion about the issue. They are able to capitalize on different sources of information and can proactively enter into a dialogue with the decision-makers on issues of interest to them.

Since ICT-enabled governance models can directly connect individuals with officials in the government and decision-makers, the impact is immediate, and puts greater assess and control over governance mechanism in the

hands of individuals. Spaces therefore get created within the existing governance mechanisms that would be democratically informed by citizen's voice. This process makes the government more responsible and accountable to its citizens.

Digital governance models bring with them the potential to bring in far-reaching changes in governance structures in developing countries. The approach to it however has to be cautious and well thought out. Simple replication of popular e-governance models of the developed world in the developing world will not be effective and may end up marginalizing people who are on the wrong side of the digital divide. Local needs and priorities should guide selection and nature of implementation of e-governance models in an evolutionary manner.

Creativity lies in the use of information and blending it with the potential offered by ICT to create customized low-cost digital governance models. Solutions may often emerge by analysis of reasons behind governance failures by the people themselves and then imaginatively using digital governance models to focus on the weakest spot to bring gradual improvements in governance. Small businesses are significant landmarks in the path of transformation to good governance.